Haleh Damirchi

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Machine learning researcher and PhD graduate from Queen's University with expertise in deep learning, probabilistic forecasting, and pedestrian trajectory prediction.

Work Experience

Graduate Research Fellow — May 2021 – April 2026
Queen's University — *Aiim lab, RCV lab*

- Developed deep learning methods for pedestrian trajectory forecasting and crowd counting.
- Published peer-reviewed research at leading venues including ICASSP and WACV.
- Communicated technical findings to research collaborators and translated model-performance analysis into iterative improvements to experiments.

University Lecturer — Jan 2026 – April 2026
Queen's University — *Introduction to Data Science*

- Delivered weekly lectures and provided responsive academic support to a large undergraduate class.
- Led lab sessions and guided students through assignments, projects, and core data science concepts.
- Developed lecture materials and designed assessments aligned with course learning objectives.

Teaching Assistant — 2021 – 2025
Queen's University

- Assisted with Machine Vision with Deep Learning, C Programming, and Artificial Intelligence courses.
- Evaluated assignments and exams with careful attention to consistency and academic standards.
- Mentored students through coursework and helped them strengthen debugging and problem-solving skills.

Teaching Assistant — 2019 – 2021
Amirkabir University of Technology

- Assisted in teaching Machine Learning and Logic Circuits courses for undergraduate students.
- Designed tutorials, quizzes, and exam materials to reinforce core concepts and improve student understanding.
- Conducted weekly tutorial sessions and provided clear technical guidance in problem solving and coursework.

R&D Intern — Summer, Fall 2016
Aria Kavosh Industrial Corp.

- Built and wired an educational control unit integrating relays, PLCs, and an HMI interface.
- Developed detailed wiring and logic diagrams in Microsoft Visio to document system design.

R&D Intern — Summer 2014
Tabriz Peguh

- Conducted research on a sensor-based system for distinguishing between gasoline and diesel fuel in vehicle fuel tanks.

Technical Skills

Programming: Python, C, SQL, MATLAB
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn
Architectures and Methods: CNNs, RNNs, Transformers, probabilistic forecasting, multimodal learning
Tools: Git, LaTeX, MySQL

Publications

[1] **H. Damirchi**, M. Greenspan and A. Etemad, "A Comprehensive and Critical Analysis of Metrics and Evaluation Practices in Pedestrian Trajectory Forecasting", Under Review.

[2] **H. Damirchi**, A. Etemad and M. Greenspan, "Socially-informed Reconstruction for Pedestrian Trajectory Forecasting", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).

[3] **H. Damirchi**, M. Greenspan and A. Etemad, "Context-Aware Pedestrian Trajectory Prediction with Multimodal Transformer", International Conference on Image Processing (ICIP 2023).

[4] M. Zand, **H. Damirchi** and A. Farley, M. Molahasani, "Multiscale Crowd Counting and Localization by Multitask Point Supervision ", IEEE International Conference on Acoustics (ICASSP 2022).

[5] **H. Damirchi**, S. Seyedin and S. M. Ahadi, "Speaker Extraction Using Stacked BLSTM Optimized with Frequency-domain Differentiated Spectrum Loss", Iranian Conference on Electrical Engineering (ICEE 2020).

[6] **H. Damirchi**, S. Seyedin, S. M. Ahadi, "Improving the Loss Function Efficiency for Speaker Extraction Using Psychoacoustic Effects", *Applied Acoustics*.

Education

Queen's University — Kingston, Canada
PhD. Electrical and Computer Engineering — *2021 – 2026*
Thesis: Pedestrian Trajectory Forecasting in Urban Scenarios

Amirkabir University of Technology — Tehran, Iran
M.Sc. Electrical Engineering — *2017 – 2020*
Thesis: Single-channel Speaker Extraction based on Deep Learning

Tabriz University — Tabriz, Iran
B.Sc. Electrical Engineering — *2012 – 2016*

Selected Projects

Trajectory Forecasting | *Pytorch*

- Developed multimodal forecasting models that predict multiple plausible pedestrian trajectories from observed motion and contextual information. |
- Implemented and compared CNN, Recurrent, graph-based, and Transformer architectures on ETH/UCY benchmark, PIE and JAAD datasets.
- Analyzed evaluation methods for trajectory forecasting both theoretically and empirically to encourage a comprehensive comparison between solutions.

SlouchLess | *Python*

- Packaged and published standalone Windows and Linux releases for an application that detects prolonged slouching from webcam video and provides real-time posture reminders.
- Built a data collection and training pipeline that learns posture patterns.

Crowd Counting | *Pytorch*

- Developed a multitask network for simultaneous crowd counting and point-based person localization.
- Used pretrained VGG features and point-level supervision to jointly optimize density estimation and localization objectives.

Speaker Extraction | *TensorFlow, Pytorch, Matlab*

- Developed stacked BLSTM models for extracting a target speaker from single-channel speech mixtures.
- Designed frequency-aware and psychoacoustic loss functions using Mel and Gammatone filter banks and perceptual hearing models.

Selected Projects

Speech Recognition and Enhancement | *Matlab*

- Used Hidden Markov Models (HMM) and Dynamic Time Warping (DTW) to recognize an utterance.
- Enhanced the input noisy speech signal using MMSE and Spectral Subtraction algorithm.
- Applied Ramirez04 algorithm to detect the activity of speech signals.

Backpropagation using Evolutionary Algorithms| *Python*

- Formulated a deep neural network training approach based on Gray Wolf and Particle Swarm Optimization.

Service and Professional Activities

- **Reviewer**: ICLR, NeurIPS 2026; NeurIPS, AISTATS, ICLR, and ICML 2025; NeurIPS and ICASSP 2024; IEEE Transactions on Artificial Intelligence; IEEE Transactions on Affective Computing
- **Guest Lecturer**: Computer Vision with Deep Learning, Queen's University, Fall 2025
- **Organizing Committee Member**: International Conference on Signal Processing and Intelligent Systems, Amirkabir University of Technology, Dec. 2018

Languages

- Azeri, Persian: Native
- English: Fluent